

MARIUPOL', UkraineWMO: **347120**Lat: **47.0425N**Lon: **37.4844E**Elev: **70**StdP: **100.49**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.8	-12.9	-19.4	0.7	-14.6	-16.6	0.9	-11.8	14.3	-6.6	13.3	-5.5	5.6	50	0.584

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	32.2	21.4	30.7	21.2	29.2	20.8	23.9	28.9	22.9	28.1	22.0	27.3	4.3	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.3	17.1	26.8	21.1	15.9	26.0	20.1	14.9	25.2	72.0	29.0	68.1	28.1	64.8	27.3	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	12.8	11.1	9.7	-18.3	34.3	4.5	2.3	-21.6	35.9	-24.2	37.2	-26.8	38.5	-30.1	40.2	
(5)				-18.7	25.7	4.4	1.1	-21.9	26.6	-24.5	27.2	-26.9	27.9	-30.1	28.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.3	-2.7	-1.8	2.9	9.9	16.5	21.0	23.6	23.1	17.2	10.3	3.9	-0.7
	DBStd	10.27	5.16	4.85	3.55	3.52	3.83	3.35	2.86	3.41	3.79	4.32	4.31	4.80
	HDD10.0	1559	394	331	219	43	2	0	0	0	2	49	185	333
	HDD18.3	3383	653	564	477	253	83	13	1	3	65	249	432	591
	CDD10.0	1676	0	0	0	41	202	329	420	405	216	59	3	0
	CDD18.3	460	0	0	0	1	25	92	162	149	29	1	0	0
	CDH23.3	3842	0	0	0	5	113	688	1509	1407	119	1	0	0
	CDH26.7	1132	0	0	0	1	12	162	473	470	15	0	0	0
Wind	WSAvg	4.6	5.1	5.2	5.2	4.9	4.2	4.1	4.0	4.2	4.5	4.5	4.9	5.0
Precipitation	PrecAvg	525	52	39	40	41	46	57	42	33	45	38	46	50
	PrecMax	852	193	133	85	134	128	133	166	108	185	85	99	118
	PrecMin	378	13	9	6	2	1	8	2	0	4	11	4	9
	PrecStd	120	36	28	20	28	32	32	36	33	39	19	28	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.9	8.1	12.9	22.7	28.1	32.0	34.1	35.6	28.2	22.2	14.7	9.4
		MCWB	5.6	6.1	8.5	15.1	19.1	21.2	21.2	20.4	18.9	15.6	12.7	8.5
	2%	DB	5.2	6.1	10.8	18.9	26.0	30.0	32.1	32.8	26.0	19.9	12.6	7.3
		MCWB	4.5	4.7	7.0	12.5	18.5	20.8	21.8	21.4	18.7	15.2	11.1	6.5
	5%	DB	4.0	4.6	9.1	16.9	24.5	28.6	30.7	30.9	24.5	18.2	11.0	6.1
		MCWB	3.5	3.5	6.3	12.0	17.6	20.1	21.6	20.8	17.9	14.8	9.8	5.6
	10%	DB	2.8	3.4	7.8	15.3	23.0	27.0	29.2	29.2	23.2	16.8	9.8	5.0
		MCWB	2.4	2.7	5.6	11.2	17.1	19.8	21.1	20.8	17.0	14.2	8.8	4.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	6.8	9.4	15.5	21.3	24.3	25.1	25.4	22.3	18.2	13.2	8.8
		MCDB	6.5	7.6	12.1	20.7	25.6	28.3	30.1	30.2	25.8	19.8	14.0	9.1
	2%	WB	4.7	5.0	7.7	13.7	19.9	22.8	24.0	23.7	20.5	16.5	11.5	6.7
		MCDB	5.0	5.7	9.6	17.2	23.8	27.1	29.5	28.8	24.2	18.6	12.3	7.2
	5%	WB	3.5	3.7	6.7	12.7	18.8	21.8	23.0	22.6	19.4	15.4	10.1	5.6
		MCDB	3.9	4.3	8.3	15.8	23.0	26.4	28.5	28.2	22.7	17.4	10.9	6.0
	10%	WB	2.4	2.8	5.8	11.8	17.7	20.8	22.1	21.7	18.3	14.3	8.9	4.6
		MCDB	2.7	3.3	7.3	14.5	21.8	25.4	27.6	27.3	21.7	16.4	9.5	5.0
Mean Daily Temperature Range	5% DB	MDBR	4.4	4.9	5.5	6.9	7.6	8.3	8.8	8.9	8.0	6.5	4.9	4.4
		MCDBR	4.4	5.8	7.7	9.5	9.3	10.1	9.9	10.5	9.1	7.8	6.5	4.9
		MCWBR	4.0	4.7	5.1	5.8	5.7	6.2	5.9	6.3	6.0	5.3	5.3	4.5
	5% WB	MCDBR	4.1	5.2	6.6	7.8	8.0	8.6	8.6	8.9	7.6	7.0	5.9	4.7
		MCWBR	3.9	4.5	4.9	5.2	5.4	5.8	5.9	6.1	5.9	5.3	5.2	4.4
Clear-Sky Solar Irradiance	taub	0.345	0.376	0.407	0.455	0.418	0.420	0.445	0.444	0.414	0.384	0.356	0.346	
	taud	2.277	2.200	2.174	2.084	2.298	2.325	2.258	2.246	2.297	2.335	2.332	2.283	
	Ebn at Noon	715	767	803	798	849	847	818	802	791	754	698	663	
	Edh at Noon	80	105	126	151	126	124	131	127	111	92	76	71	
All-Sky Solar Radiation	RadAvg	0.97	1.89	3.04	4.72	6.14	6.48	6.45	5.79	4.14	2.47	1.20	0.77	
	RadStd	0.16	0.33	0.37	0.44	0.50	0.43	0.34	0.38	0.32	0.33	0.18	0.12	